

1)Connect to hadoop

```
pict@pict-OptiPlex-3020:~$ ssh hdoop@10.10.14.149
```

```
hdoop@10.10.14.149's password:
```

2)check directories

```
hdoop@pict-OptiPlex-3020:~$ ls
```

```
31331
```

```
31338
```

```
31343
```

```
31356
```

```
31406_weather_project
```

```
31410
```

```
31413
```

3)change directory

```
hdoop@pict-OptiPlex-3020:~$ cd $HADOOP_HOME/sbin
```

```
hdoop@pict-OptiPlex-3020:~/hadoop-3.4.0/sbin$
```

4)Create folder 31360

```
hdoop@pict-OptiPlex-3020:~/hadoop-3.4.0/sbin$ mkdir 31360
```

5)Go to 31360

```
hdoop@pict-OptiPlex-3020:~/hadoop-3.4.0/sbin$ cd 31360
```

```
hdoop@pict-OptiPlex-3020:~/hadoop-3.4.0/sbin/31360$
```

6)Create data.txt file

```
hdoop@pict-OptiPlex-3020:~/hadoop-3.4.0/sbin/31360$ nano data.txt
```

DATA: Apache Hadoop is an open-source framework used to securely store, process, and analyze massive datasets (from gigabytes to petabytes) across clusters of commodity hardware. It enables distributed computing, breaking large workloads into smaller tasks to run in parallel, making it highly scalable and fault-tolerant

7)Create WordCount java program:

```
nano WordCount.java
```

```
import java.io.IOException;
import java.util.StringTokenizer;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
```

```

import org.apache.hadoop.mapreduce.Reducer;
import
org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import
org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;
public class WordCount {
public static class TokenizerMapper
extends Mapper<Object, Text, Text, IntWritable>{
private final static IntWritable one = new
IntWritable(1);
private Text word = new Text();
public void map(Object key, Text value, Context context)
throws IOException, InterruptedException {
StringTokenizer itr = new
StringTokenizer(value.toString());
while (itr.hasMoreTokens()) {
word.set(itr.nextToken());
context.write(word, one);

```

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```

}
}
}

```

```

public static class IntSumReducer
extends Reducer<Text,IntWritable,Text,IntWritable> {
private IntWritable result = new IntWritable();
public void reduce(Text key, Iterable<IntWritable>
values,

```

Context context) throws IOException,

```

InterruptedException {
int sum = 0;
for (IntWritable val : values) {
sum += val.get();
}
result.set(sum);
context.write(key, result);
}
}
public static void main(String[] args) throws Exception {
Configuration conf = new Configuration();
Job job = Job.getInstance(conf, "word count");
job.setJarByClass(WordCount.class);

```

```
job.setMapperClass(TokenizerMapper.class);
job.setReducerClass(IntSumReducer.class);
job.setOutputKeyClass(Text.class);
job.setOutputValueClass(IntWritable.class);
```

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```
FileInputFormat.addInputPath(job, new Path(args[0]));
FileOutputFormat.setOutputPath(job, new Path(args[1]));
System.exit(job.waitForCompletion(true) ? 0 : 1);
}
}
```

8 Compile the file

```
hadoop@pict-OptiPlex-3020:~/hadoop-3.4.0/sbin/31360$ javac -classpath `hadoop classpath`
WordCount.java
```

Create jar file

3) check haddop process

localhost: nodemanager is running as process 22038. Stop it first and ensure
/tmp/hadoop-hdoo-nodemanager.pid file is empty before retry.

```
hadoop@pict-OptiPlex-3020:~/hadoop-3.4.0/sbin/31360$ jps
```

21904 ResourceManager

24020 NameNode

25861 Jps

22038 NodeManager

10935 MainGenericRunner

24200 DataNode

13243 Main

24412 SecondaryNameNode

```
hadoop@pict-OptiPlex-3020:~/hadoop-3.4.0/sbin/31360$ hdfs mkdir hdfs://localhost:9000/31360
```

```
hadoop@pict-OptiPlex-3020:~/hadoop-3.4.0/sbin/31360$ hdfs dfs -ls /31360/output
```

Found 2 items

```
-rw-r--r-- 1 hadoop supergroup 0 2026-04-09 11:52 /31360/output/_SUCCESS
```

```
-rw-r--r-- 1 hadoop supergroup 388 2026-04-09 11:52 /31360/output/part-r-00000
```

```
hadoop@pict-OptiPlex-3020:~/hadoop-3.4.0/sbin/31360$ hdfs dfs -cat
```

```
/31360/output/part-r-00000
```

(from 1

Apache 1

Hadoop 1

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hardware.	1
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into	1
is	1
it	1
large	1
making	1
massive	1
of	1
open-source	1
parallel,	1
petabytes)	1
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run	1
scalable	1
securely	1
smaller	1
store,	1
tasks	1
to	3
used	1
workloads	1

hadoop@pict-OptiPlex-3020:~/hadoop-3.4.0/sbin/31360\$

```
Activities Terminal Apr 9 11:57
hadoop@pict-OptiPlex-3020: ~/hadoop-3.4.0/sbin/31360

tnpdata docs.google.com/document/d/1sMcj0emg7iabmCXzllZ_p1mkZXFh4GIV1EtjG-cCxQ/edit?pli=1&tab=t.0
Videos
weather1.jar
WeatherAnalysis$WeatherMapper.class
WeatherAnalysis$WeatherReducer.class
WeatherAnalysis.class
weatherAnalysis.jar
WeatherAnalysis.jar
WeatherAnalysis.java
WeatherAverage$WeatherMapper.class
WeatherAverage$WeatherReducer.class
WeatherAverage.class
WeatherAverage.java
WeatherAvg.java
weatherclean.jar
weather_clean.txt
weather.csv
weather_input
weather.jar
weather_mapper
weather_mr you add to the document
weather_project
weather_simple.txt
weather.txt
WordCount$IntSumReducer.class
WordCount$TokenizerMapper.class
WordCount.class
wordcount.jar
WordCount.jar
WordCount.java
WordCount.java

hadoop@pict-OptiPlex-3020:~$ cd $HADOOP_HOME/sbin
hadoop@pict-OptiPlex-3020:~/hadoop-3.4.0/sbin$ ls
31303 31329 31409 start-dfs.sh
31304 31330 31410 start-secure-dns.sh
31305 31332 31411 start-yarn.cmd
31306 31333 31412 start-yarn.sh
```

```
Activities Terminal Apr 9 11:57
hadoop@pict-OptiPlex-3020: ~/hadoop-3.4.0/sbin/31360

Caused by: java.io.IOException: Input path does not exist: hdfs://localhost:9000/31248/input
at org.apache.hadoop.mapreduce.lib.input.FileInputFormat.singleThreadedListStatus(FileInputFormat.java:315)
... 19 more
hadoop@pict-OptiPlex-3020:~/hadoop-3.4.0/sbin/31360$ hadoop jar WordCount.jar WordCount /31360/input /31360/output
2026-04-09 11:51:56,518 INFO client.DefaultHadoopFailoverProxyProvider: Connecting to ResourceManager at /127.0.0.1:8032
2026-04-09 11:51:56,737 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-04-09 11:51:56,766 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/hadoop/.staging/job_1775715151830_0008
2026-04-09 11:51:57,075 INFO InputFileInputFormat: Total Input files to process : 1
2026-04-09 11:51:57,972 INFO mapreduce.JobSubmitter: number of splits:1
2026-04-09 11:51:58,147 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1775715151830_0008
2026-04-09 11:51:58,147 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-04-09 11:51:58,265 INFO conf.Configuration: resource-types.xml not found
2026-04-09 11:51:58,265 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-04-09 11:51:58,314 INFO impl.YarnClientImpl: Submitted application application_1775715151830_0008
2026-04-09 11:51:58,354 INFO mapreduce.Job: The url to track the job: http://pict-OptiPlex-3020:8088/proxy/application_1775715151830_0008/
2026-04-09 11:51:58,355 INFO mapreduce.Job: Running job: job_1775715151830_0008
2026-04-09 11:52:03,452 INFO mapreduce.Job: Job job_1775715151830_0008 running in uber mode : false
2026-04-09 11:52:03,454 INFO mapreduce.Job: map 0% reduce 0%
2026-04-09 11:52:07,555 INFO mapreduce.Job: map 100% reduce 0%
2026-04-09 11:52:13,596 INFO mapreduce.Job: map 100% reduce 100%
2026-04-09 11:52:14,625 INFO mapreduce.Job: Job job_1775715151830_0008 completed successfully
2026-04-09 11:52:14,789 INFO mapreduce.Job: Counters: 54

File System Counters
  FILE: Number of bytes read=586
  FILE: Number of bytes written=618625
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=423
  HDFS: Number of bytes written=388
  HDFS: Number of read operations=8
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=2
  HDFS: Number of bytes read erasure-coded=0

Job Counters
  Launched map tasks=1
  Finished map tasks=1
```

```
Activities Terminal Apr 9 11:56
hdoop@pict-OptiPlex-3020: ~/hadoop-3.4.0/sbin/31360
across 1
an 1
analyze 1
and 2
breaking 1
clusters 1
commodity 1
computing 1
datasets 1
distributed 1
enables 1
fault-tolerant 1
framework 1
gigabytes 1
hardware 1
highly 1
in 1
into 1
is 1
it 1
large 1
making 1
massive 1
of 1
open-source 1
parallel 1
petabytes 1
process 1
run 1
scalable 1
securely 1
smaller 1
store 1
tasks 1
to 3
used 1
workloads 1
hdoop@pict-OptiPlex-3020:~/hadoop-3.4.0/sbin/31360$

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